Øptimus Data Science Predictive Modeling Exercise

Who will vote in Nevada in November 2014?

A turnout prediction for 48,859 registered voters, built as if it were May 2014

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The question and the data

Nov 2014

Predict who turns out in the general election, using only what was known in May 2014.

48,859

unique Nevada voters. The file had 50,000 rows; 1,141 exact duplicate rows were removed.

39

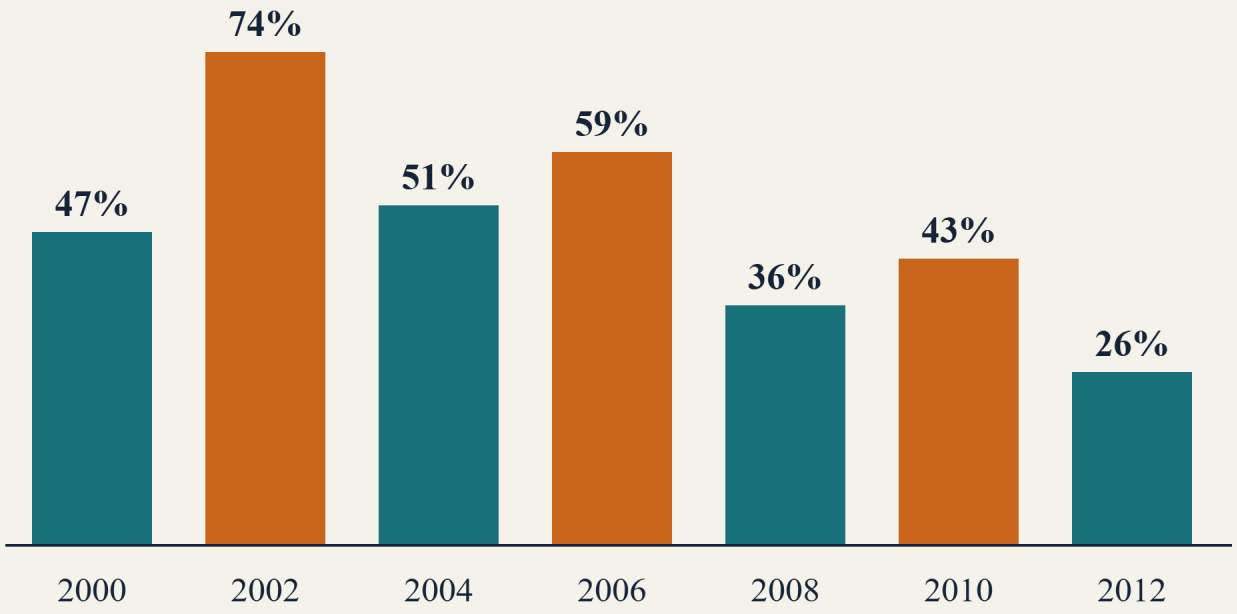
starting variables: age, party, voting history and the turnout of each voter's precinct.

Key rule: the 2014 primary result is not used, because it would not exist yet in May 2014.

Source: Nevada voter file (voterfile.csv)

What the voting history shows

Share of voters who voted in each general election



Orange bars are midterm years, teal bars are presidential years.

Turnout drops across the file

From 74% in 2002 to 26% in 2012, so predictions are best read as a ranking of voters.

Five fields are almost empty

Donor, NASCAR, dog-owner and instrument fields are over 90% blank, so they were dropped.

Source: voter-history columns in the Nevada voter file

How the prediction works

1. Learn from the last midterm

The 2010 general election has a known result, so the model learns from it using only what was known before 2010.

2. Describe voters by "elections ago"

Past votes 2, 4, 6, 8 and 10 years back, plus precinct turnout, age and profile. The same rules then apply to 2014.

3. Gradient boosting model

Compared with a simpler logistic regression and tuned with 5-fold cross-validation. Boosting scored higher.

4. Turn scores into votes

Each voter gets a probability from 0 to 1. A voter is predicted to vote when it reaches 0.42, the cut-off that best balanced false alarms and misses.

Method: HistGradientBoostingClassifier with probability calibration

How accurate is it?

0.92
AUC score. 1.0 is perfect and 0.5 is a coin flip.

84%
of voters classified correctly.

84%
of people who really voted were found.

81%
of people predicted to vote really did.

Model (5-fold cross-validation)	AUC	Brier error (lower is better)
Logistic regression	0.912	0.114
Gradient boosting	0.923	0.108

Top figures: 20% of 2010 voters held back and never used for training or tuning

What we found

37.5%

of voters are predicted to turn out in November 2014.

Past behavior and age lead

Voting in 2012, then age, then voting in 2010 matter most. Profile details add little.

Most scores are decisive

A quarter of voters score below 7% and a quarter score above 83%.

Use it as a ranking

Turnout fell across elections in this file, so rank voters by probability instead of trusting the exact total.

Predictions: vote and vote_prob for every voter in predictions.csv